

Poisson distribution and Extended Poisson Model

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1 Data

Table 1: *Ceriodaphnia dubia* data.

Concentration (%)	Number of young	Mean	Variance
0	27, 30, 29, 31, 16, 15, 18, 17, 14, 27	22.4	48.044
1.5625	32, 35, 32, 26, 18, 29, 27, 16, 35, 13	26.3	64.011
3.125	39, 30, 33, 33, 36, 33, 33, 27, 38, 44	34.6	23.378
6.25	27, 34, 36, 34, 31, 27, 33, 31, 33, 31	31.7	8.678
12.5	10, 13, 7, 7, 7, 10, 10, 16, 12, 2	9.4	15.156

2 Poisson distribution, Negative binomial distribution and Generalized Waring distribution

Ignore the regressors and we here compare the Poisson distribution, Negative binomial distribution and Generalized Waring distribution. Negative binomial distribution and GWD solve the overdispersion problem.

Negative binomial distribution $Y \sim NB(r, p)$ has a probability mass function as

$$P(Y = k) = \binom{r+k-1}{k} \left(\frac{r}{r+\mu}\right)^r \left(\frac{\mu}{r+\mu}\right)^k, \quad k = 0, 1, 2, \dots$$

and can be regarded as a mixture Poisson model.

$$\begin{aligned} Y|\lambda &\sim \text{Poisson}(\lambda) \\ \lambda &\sim \text{Gamma}(\alpha, \beta) \end{aligned}$$

As a result,

$$\begin{aligned} p &= \frac{\beta}{\beta+1} \\ r &= \alpha \end{aligned}$$

$$\text{Var}(Y) = E[Y] + E[Y]^2/r > E[Y]$$

Generalized Waring distribution $Y \sim GWD(a, k, \rho)$ has a probability mass function as

$$P(Y = y) = \frac{(\rho)_k (a)_y (k)_y}{(a + \rho)_{y+k} y!}, \quad y = 0, 1, 2, \dots$$

$GWD(a, k, \rho)$ is the result of a mixture Poisson-Gamma-Beta Model.

$$\begin{aligned} Y|\lambda &\sim \text{Poisson}(\lambda) \\ \lambda|p &\sim \text{Gamma}(k, \frac{1-p}{p}) \\ p &\sim \text{Beta}(a, \rho) \end{aligned}$$

When $\rho > 2$, the 2nd moment is finite.

$$\begin{aligned} E[Y] &= \frac{ak}{\rho - 1} \\ \text{Var}(Y) &= \frac{ak(a + \rho - 1)(k + \rho - 1)}{(\rho - 1)^2(\rho - 2)} \end{aligned}$$

The relationship between mean and variance is,

$$\boxed{\text{Var}(Y) > E[Y]}$$

3 Extended Poisson Model

The above three distributions cannot be compatible with covariate-dependent under/over-dispersion. Actually, NB and GWD only give overdispersion and cannot deal with under-dispersion. To model the covariate-dependent under/over-dispersion, we are about to use Extended Poisson Process Models.

Extended Poisson Process Models (EPPMs) give rise to discrete probability distributions on $0, 1, 2, \dots$ of the form, for arbitrary n ,

$$[p_0 \ p_1 \ p_2 \ \cdots \ p_n] = [1 \ 0 \ 0 \ \cdots \ 0] \exp(Q), \quad (1)$$

where the matrix Q is given by

$$Q = \begin{pmatrix} -\lambda_0 & \lambda_0 & 0 & \cdots & 0 \\ 0 & -\lambda_1 & \lambda_1 & \cdots & 0 \\ 0 & 0 & -\lambda_2 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \lambda_{n-1} \\ 0 & 0 & 0 & \cdots & -\lambda_n \end{pmatrix} \quad (2)$$

for a sequence $\lambda_0, \lambda_1, \lambda_2, \dots$ of positive parameters. **Constant λ 's give rise to the Poisson distribution, with increasing λ 's resulting in over-dispersion relative to the Poisson distribution and decreasing λ 's under-dispersion.**

A fully parametric form for λ_i is used, provided by the three-parameter function

$$\lambda_i = a(b + i)^c, \quad i = 0, 1, 2, \dots, \quad (3)$$

with $a > 0$, $b > 0$ and $c \leq 1$, which gives rise to a broad family of discrete distributions **admitting under-dispersion relative to the Poisson distribution ($c < 0$), the Poisson distribution itself ($c = 0$), and over-dispersion relative to the Poisson distribution ($c > 0$) including the negative binomial distribution ($c = 1$).** The restriction $c \leq 1$ in Equation (3) is necessary for a proper probability distribution.

An **approximation to the mean** is

$$m = b \left[\left\{ 1 + \frac{a(1-c)}{b^{1-c}} \right\}^{1/(1-c)} - 1 \right], \quad (4)$$

so that covariate (x , say) dependence in the mean can be incorporated by modelling m as a function of the usual linear predictor $x^\top \beta$ with the parameter a given by

$$a = \begin{cases} \frac{(m+b)^{1-c} - b^{1-c}}{1-c}, & c < 1, \\ \log \left(1 + \frac{m}{b} \right), & c = 1. \end{cases} \quad (5)$$

An **approximation to the variance** of the distribution defined by Equations (1)–(3) is

$$v = \frac{b(m/b + 1) [1 - (m/b + 1)^{2c-1}]}{1 - 2c}, \quad (6)$$

so that additional covariate dependence in the variance can be incorporated by having v as a function of another linear predictor and then solving the equation

$$\frac{v}{m+b} = \frac{\exp\{(2c-1)\log(m/b+1)\} - 1}{2c-1} \quad (7)$$

for c in terms of v , m and b .

The algorithm for the fitting, we set $\theta = (\beta_m, \beta_v, \eta := \log b)$ and

$$\begin{aligned} \log m &= \beta_m \cdot x_m \\ \log v &= \beta_v \cdot x_v \end{aligned}$$

where x_m and x_v are covariates for the mean and variance. With known m, v, b , we solve c using equation (7) and then with known m, v, b, c we solve a using equation (5). After that, using a, b, c , we compute the λ 's so that we can calculate the likelihood $p(Y = y | \beta_m, \beta_v, \eta)$.

Maximization of the log likelihood gives us the MLE estimates of β_m, β_v, η .

Tips for the algorithm

1. Solve for c in a transformed variable, not directly. Direct Newton–Raphson on (7) is numerically unstable near $c = \frac{1}{2}$ (the $1 - 2c$ denominator) and near $c = 1$. Substitute

$$x = (2c-1) \log \left(\frac{m}{b} + 1 \right) =: (2c-1) L, \quad g(x) = \frac{e^x - 1}{x}, \quad (8)$$

so that (7) becomes $g(x) = t$ with

$$t = \frac{\nu}{L(m+b)}, \quad (9)$$

and start the iteration at $x_0 = L$ (which corresponds to $c = 1$). Since g is convex increasing, Newton–Raphson converges monotonically from the right with no overshoot, and

$$c = \frac{1}{2} \left(\frac{x}{L} + 1 \right). \quad (10)$$

(This is precisely what the appendix code does.)

2. Enforce admissibility — this is the crux. The model is proper *iff* $c \leq 1$, and

$$c \leq 1 \iff \frac{\nu}{m+b} \leq \frac{m}{b} \iff \nu \leq \frac{m(m+b)}{b}. \quad (11)$$

At $c = 0$ this gives $\nu = m$ (Poisson), and at $c = 1$ it gives the negative binomial variance $\nu = m + \frac{m^2}{b}$ (with $r = b$). If the fitted ν exceeds $m(m+b)/b$, the indicated $c > 1$ is *inadmissible* and you must return a large penalty (or $-\infty$ log-likelihood) so that the optimizer is pushed back into the admissible region. The appendix code does this by setting all probabilities to **1e-8**.

3. Handle the $c \rightarrow 1$ boundary. Use the logarithmic form

$$a = \log\left(1 + \frac{m}{b}\right) \quad (12)$$

when c is close to 1 (threshold, e.g., $1 - 10^{-8}$); otherwise (5) is a 0/0 form.

4. m and ν are only *approximations*. Equations (4) and (6) are approximate moments; the true mean and variance of the fitted law must be recomputed numerically from p after fitting. Hence β_m and β_ν are coefficients for the *approximate* moments — exact for the Poisson ($c = 0$) and negative binomial ($c = 1$) cases, approximate otherwise. (In the examples the approximation is excellent, but that is empirical, not guaranteed.)

5. b is a nuisance parameter and is badly identified. Its standard error is large (e.g. 2.71 and 4.43 in the two fits). Consequences:

- The objective can be flat/ill-conditioned in $\eta = \log b$.
- For strongly under-dispersed data, $\hat{b} \rightarrow \infty$ and the general $\lambda_i = a(b+i)^c$ *blows up numerically*. You must detect a large $\hat{b}$ and switch to the limiting model

$$\lambda_i = \alpha e^{\beta i}, \quad m = -\frac{\log(1 - \alpha\beta)}{\beta}, \quad \nu = \frac{e^{2\beta m} - 1}{2\beta}, \quad (13)$$

This is exactly what happened for the jejunal crypt data, where the paper had to use the limiting form.

6. The objective is non-smooth and expensive. There is a hard constraint surface ($c = 1$), a removable singularity at $c = \frac{1}{2}$, and an inner Newton–Raphson solve per observation. Expect *slow convergence* (the paper explicitly reports this for the crypt data) and difficulty for gradient-based optimizers near the boundary. Practical advice: use Nelder–Mead or BFGS with finite-difference gradients, multiple starting values, and good initial values for β_m, β_ν, η ; consider profiling over η .